SNAKE GAME

```
#include<iostream>
#include<vector>
#include<conio.h>
#include<thread>
#include<chrono>
#include<random>
#include<iomanip>

using namespace std;

int snakeX = 1;
int snakeY = 1;
int snakePrevX;
int snakePrevY;
bool alive = 1;
char head = '0';
int speed = 110;

vector<char> body(1, 'o');
vector<int> bodyX = {0};
```

```
vector<int> bodyY = {1};

string grid[16][16];

char input;
char direction = 'd';

int foodX = 8;
int foodY = 8;

char inputFunc();
void restart();

void displayMainMenu(){
    cout << "Press Any Key to Start" << endl;
    cout << "Press 1 to Exit:" << endl;
}

void moveBody(){
    for(int i = bodyX.size() - 1; i > 0; i--){
        bodyX[i] = bodyX[i - 1];
        bodyY[i] = bodyY[i - 1];
```

```
    }

    bodyX[0] = snakePrevX;
    bodyY[0] = snakePrevY;
}

void displayGrid(){
    for(int i = 0; i < 16; i++){
        for(int j = 0; j < 16; j++){

            if(i == snakeY && j == snakeX)
                cout << head << " ";

            else if(i == foodY && j == foodX)
                cout << "* ";

            else{
                int k;

                for(k = 0; k < bodyX.size(); k++){
                    if(i == bodyY[k] && j == bodyX[k]){
                        cout << body[k] << " ";
```

```
                    break;
                }
            }

            if(k == bodyX.size())
                cout << grid[i][j] << ". ";
        }
    }
    cout << endl;
  }
}

int startGame(){

  system("cls");

  while(alive == 1){

    // food spawning



    if(snakeY == foodY && snakeX == foodX){
```

```
bool onBody = true;
    while (onBody) {

    srand(time(0));
    foodX = rand() % 16;
    foodY = rand() % 16;
        bool broken = false;
    for(int z = 0; z < bodyY.size(); z++){
     if(bodyX[z] == foodX && bodyY[z] == foodY){
        broken = true;
        break;
            }
        }
        if (!broken) {
            onBody = false;
        }
    }

// size increase
    body.push_back('o');
    bodyX.push_back(bodyX.back());
    bodyY.push_back(bodyY.back());
```

```
                if(body.size() < 17){
                        speed -= 5;
                }



        }
// movement
if(_kbhit()){

    char input = _getch();

    snakePrevX = snakeX;
    snakePrevY = snakeY;

    switch(input){

        case 'w':
          if(direction != 'd'){
                    snakeY--;
                    direction = 'u';
                    moveBody();
                                }
```

```
    break;

case 'a':
  if(direction != 'r'){
            snakeX--;
            direction = 'l';
            moveBody();
                    }
    break;

case 's':
  if(direction != 'u'){
            snakeY++;
            direction = 'd';
            moveBody();
                    }
    break;

case 'd':
  if(direction != 'l'){
            snakeX++;
            direction = 'r';
```

```
            moveBody();
                    }
        break;
    }
}
else{

    snakePrevX = snakeX;
    snakePrevY = snakeY;

    switch(direction){

        case 'u':
         if(direction != 'd')
           snakeY--;
           break;

        case 'l':
         if(direction != 'r')
           snakeX--;
           break;
```

```
        case 'd':
          if(direction != 'u')
            snakeY++;
            break;

        case 'r':
          if(direction != 'l')
            snakeX++;
            break;
    }

    moveBody();
}

// checks
if(snakeX < 0 || snakeX > 15 || snakeY < 0 || snakeY > 15){
    alive = 0;
}
for(int z = 0; z < bodyY.size(); z++){
        if(bodyX[z] == snakeX && bodyY[z] == snakeY){
                alive = 0;
                }
```

```
            }
            cout << setw(18) << "Score: " << body.size() << endl;
        displayGrid();

        this_thread::sleep_for(chrono::milliseconds(speed));
        system("cls");
    }

    // ending screen

    cout << "\tGame Over :(" << endl;
    cout << "\tFinal Score: " << body.size() << endl;
    cout << "\tEnter R to Restart" << endl;
        cout << "\tEnter E to Quit: ";
        this_thread::sleep_for(chrono::milliseconds(1000));
    restart();

}
void restart(){
        char restartChoice = 'f';
    while(restartChoice != 'e' && restartChoice != 'r'){
         restartChoice = _getch();
```

```
        if(restartChoice == 'r'){
        alive = 1;
        snakeX = 1;
            snakeY = 1;
            speed = 110;
            body.clear();
            bodyX.clear();
            bodyY.clear();
            vector<char> body = {'o'};
            vector<int> bodyX = {0};
            vector<int> bodyY = {1};
            direction = 'd';
            foodX = 8;
            foodY = 8;

            startGame();
            break;
        }
        else if(restartChoice == 'e'){
            break;
        }
    }
```

```
}

int main(){

    cout << setw(18) << "Snake" << endl << endl;

    displayMainMenu();

    char choice = _getch();

    if(choice != '1'){
        startGame();
    }
    else{
        return 0;
    }
}
```